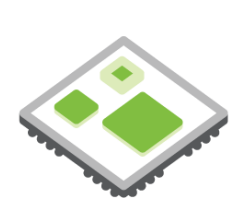
SERVER:
OPEN CHIPLET
ECONOMY



An Open Lightweight Serial Interconnect (LSI) for Composable AI Architecture

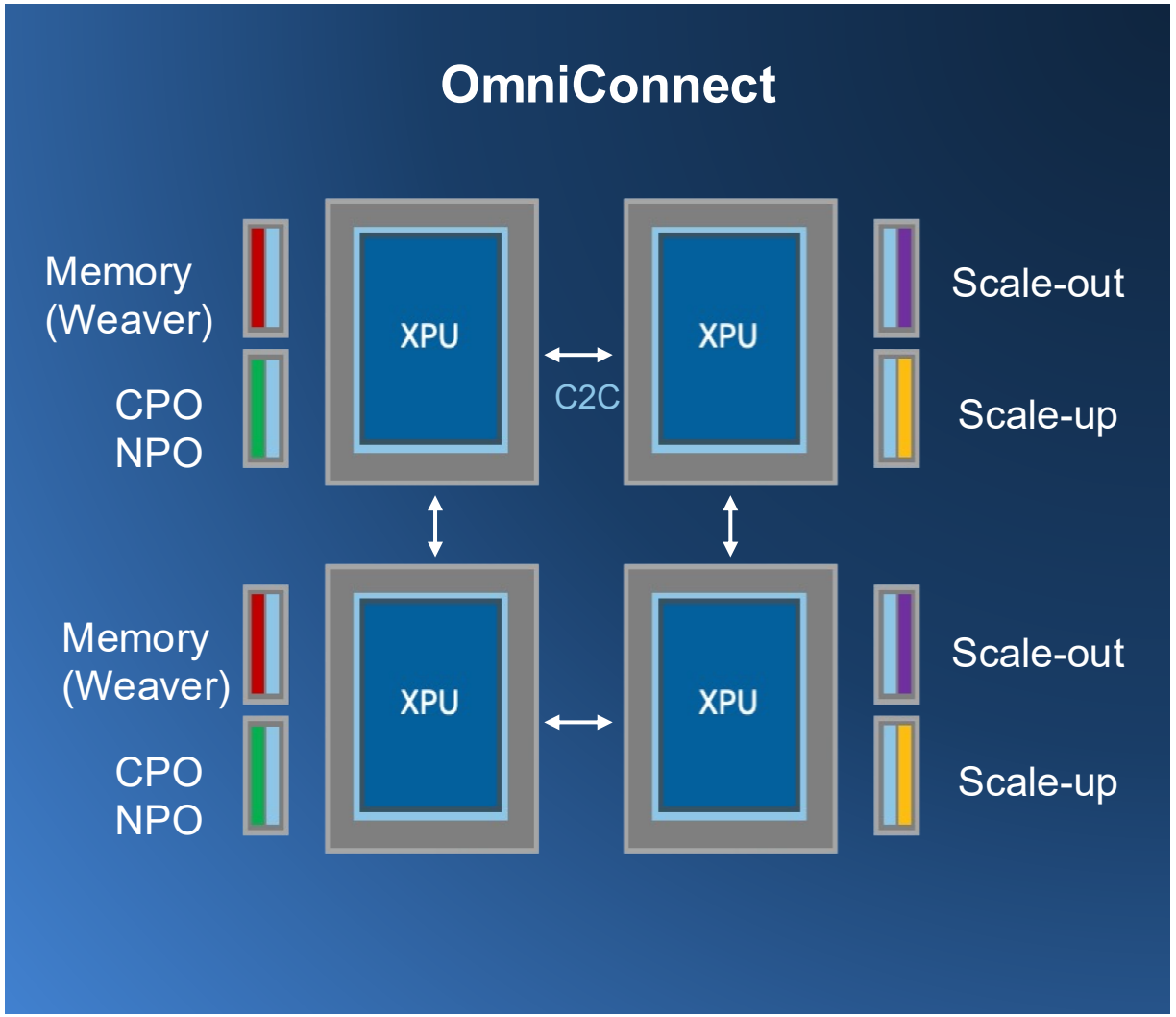
Vishal Shah, VP of Product, Credo

Allan Cantle, CEO, Nallasway

OCP PLATINUM® Member

Credo OmniConnect – A Unified I/O Interface

- A general-purpose AXI-over-VSR bus
- Chiplets and System connectivity
  - Memory (Weaver)
  - C2C (package-to-package)
  - Networking (Scale-up, Scale-out)
- Late IO binding with standard packaging
  - X% memory, y% for C2C and z% for networking
- Two Key Components
  - VSR SerDes
  - Lightweight AXI Framer



Why 112G VSR SerDes?

Right-sized for the channel: the egress-substrate gap between on-package D2D and long-reach links

On package	Off the substrate, onto the board	Backplane and cable
D2D (UCIe, BoW)	**112G VSR**	**Long reach (224G LR, PCIe/CXL)**
< 25 mm	**~10 in · 23 dB**	> 1 m · 30+ dB

	D2D	112G VSR	224G LR	PCIe 6 / CXL
Leaves the package	○	●	●	●
Beachfront density	●	●	○	○
Energy per bit	●	●	◐	◐
Latency	●	●	○	◐

● Strong fit ◐ Partial ○ Weak fit

Relative fit for the egress-substrate channel

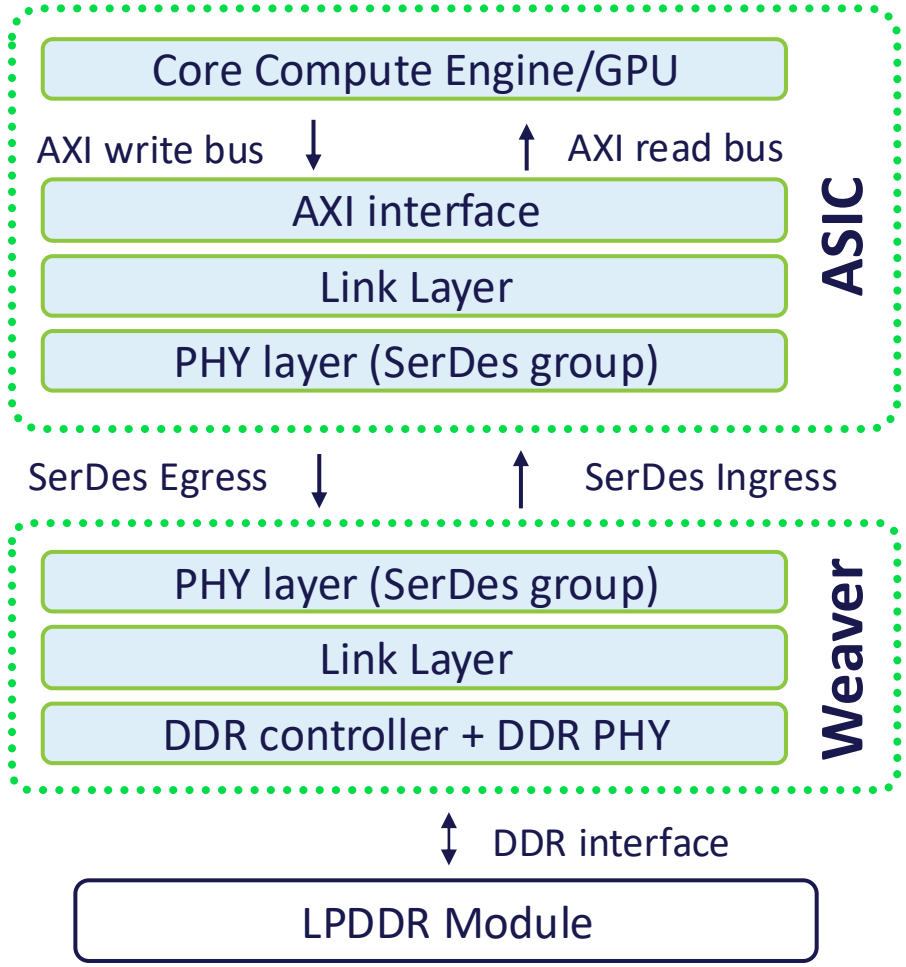
Credo 112G VSR

23 dB / 10 in	reach
< 1 pJ/bit	energy
1.8 Tb/s/mm	beachfront

Lightweight AXI Framer

- Same FLIT structure as PCIe Gen6/CXL
  - Header, CRC and FEC reused from PCIe 6 FLIT mode
- Credit based flow control
- FEC + CRC + retry for a reliable link, no software stack
- Full control over LPDDR PHYs (page status, refresh etc.)
- Designed to ensure low latency and low power over SerDes link

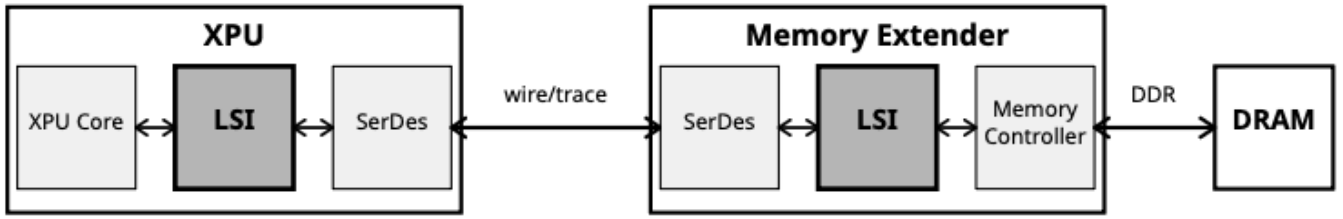
- **Foundation of the OCP LSI workstream**



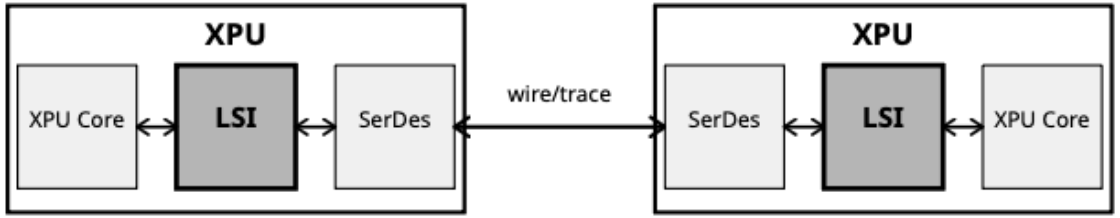
What is Lightweight Serial Interconnect (LSI)?

- Workstream initiated by Credo
- Lightweight AXI Framer + Any IEEE Compliant SerDes

XPU – to – Memory



XPU – to – XPU (C2C)



- LSI Specification leverages first generation of OmniConnect product (Weaver)

OCE LSI Workstream

- Mission Statement
  - Enable an open, low-power, modular serial interconnect ecosystem for composable chiplet-based AI architecture
- Goals
  - Open Interoperability
  - Enable modular system composition
  - Improved efficiency and scalability
  - Support system-level innovation
  - Ecosystem collaboration
- Workstream Scope
  - Phase 1 – Create foundation for open review with OCP community and initial contribution of the spec
  - Phase 2 – Identify opportunities for future enhancements, extension of the initial spec

LSI – Key Functional Blocks

TRANSACTION LAYER

AXI Extender

512-bit AXI slave → master bridge; 5 channels packetised independently

Credit Flow Control

Per-channel credits returned in DLP messages; nothing dropped

DATA LINK LAYER

FLIT Mechanism

Fixed 256 B blocks give predictable latency and buffering

Retry

ACK/NAK in every FLIT header; TX retry buffer replays

FEC

3-way interleaved ECC corrects lane bursts ≤ 17 bits

Link Training

Automatic bring-up with ordered sets; back-pressures the app while down

CRC

8-byte CRC over GF(2^8); detects up to 8 bad symbols

Scrambler

PCIe 23-bit LFSR for DC balance and lane lock

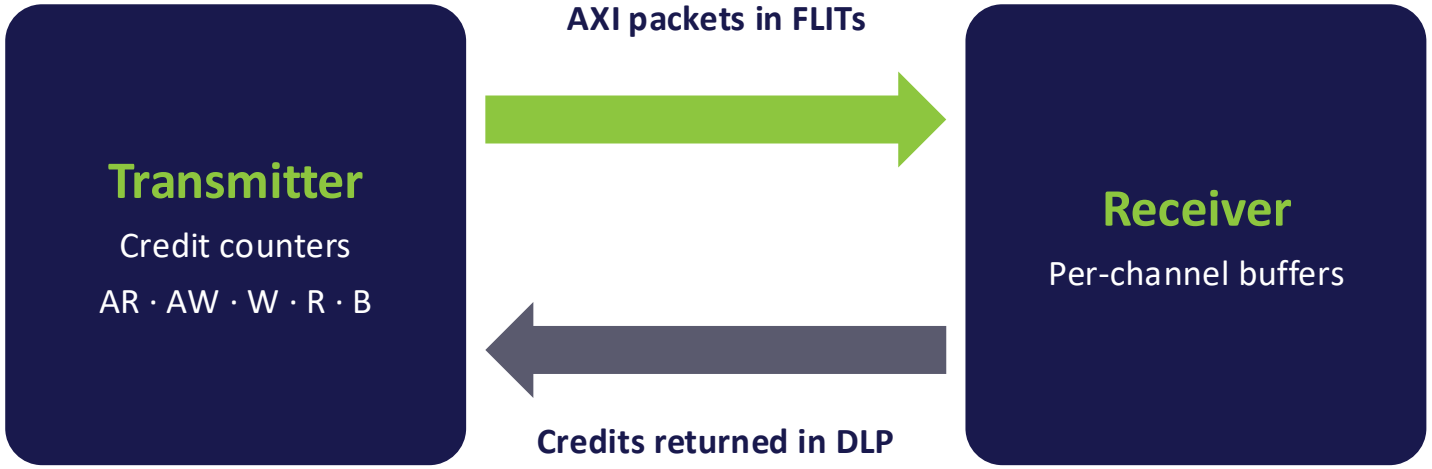
AXI Packet Formats

Type	Packet	Length	Carries
0x00	SP – Idle Space	≥ 2 B	Fills unused FLIT space
0x01	AR – Address Read	16 B	64-bit addr, 16-bit ID, len, size, burst
0x02	AW – Address Write	16 B	64-bit addr, 16-bit ID, cache, QoS, region
0x03	B – Write Response	4 B	ID + response
0x06	WH – Write Data	2 + 64·N B	1–16 beats of 512-bit data
0x08	RH – Read Data	4 + 64·N B	1–16 beats + ID and response
0x09	N – Reset / Interrupt	2 B	Inline reset and interrupt
0x0A	WT – Write w/ Strobe	74 B	One beat + 64-bit byte strobe

Packing rules

- 8-bit type header; packets padded to 2-byte boundaries
- Packed back-to-back and may cross FLIT boundaries while traffic is continuous
- Idle packet fills the rest of a sparse FLIT
- Up to 16 data beats merged per packet to cut header overhead
- Optional AXI signals (QoS, region, lock) carried as sideband

Credit-Based Flow Control



DLP format (6 bytes)

Byte	Field
0	DLP type
1	AR credits
2	AW credits
3	W credits
4	R credits
5	B credits

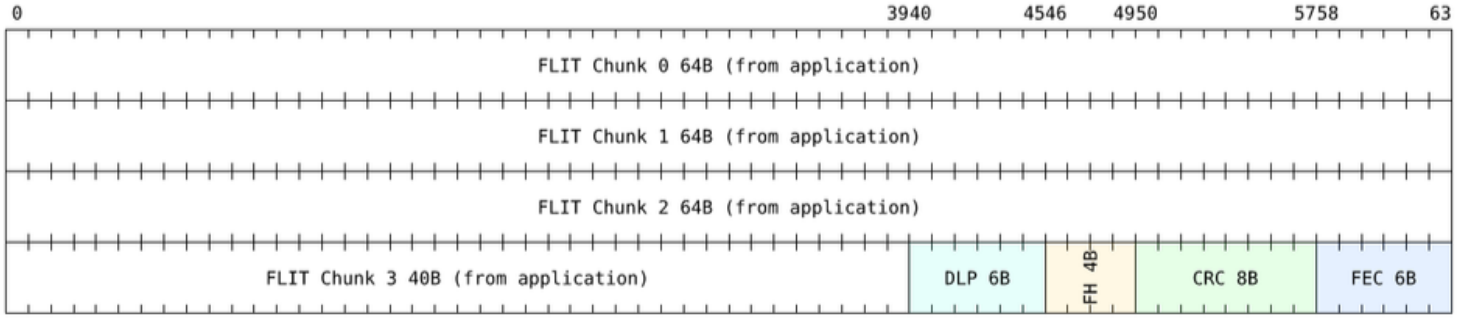
0x01 Credit Status

TX reports its current credit count; used at link init and to recover leaked credits

0x02 Credit Returned

RX releases credits as its buffers drain

FLIT Structure



Field	Bytes	Purpose
DATA	238	AXI payload in 4 chunks (64 / 64 / 64 / 46 B)
DLP	6*	Credit messages; *optional, uses chunk 3 space
FLIT Header	4	Replay sequence, ACK/NAK, idle chunks, used bytes, backpressure
CRC	8	Error detection over header + payload (242 B)
FEC	6	2 check bytes for each of 3 interleaved codewords

93%

payload efficiency: only 18 B of header, CRC and FEC per 256 B FLIT

Why fixed 256 B?

Same FLIT size as PCIe 6 / CXL 3

Fixed size keeps buffering and latency predictable

NOP FLITs keep the link busy when idle

Retry Buffer, CRC and FEC

FEC — correct first

- 3 interleaved codewords over GF(2^8), same polynomial as PCIe (0x1D)
- Single-symbol correction per codeword
- Any burst ≤ 17 bits on one lane is always corrected (up to 24 depending on alignment)
- Low overhead keeps latency low

CRC — then detect

- 8-byte CRC computed over 242 B (header + payload)
- Generator over GF(2^8) detects up to 8 bad symbols
- Runs after FEC; a failure triggers retry

Retry — then replay

- TX retry buffer holds each FLIT until ACKed
- ACK/NAK and sequence number carried in every FLIT header
- NAK makes TX replay from the failed FLIT; RX discards out-of-order FLITs
- At most one outstanding replay request

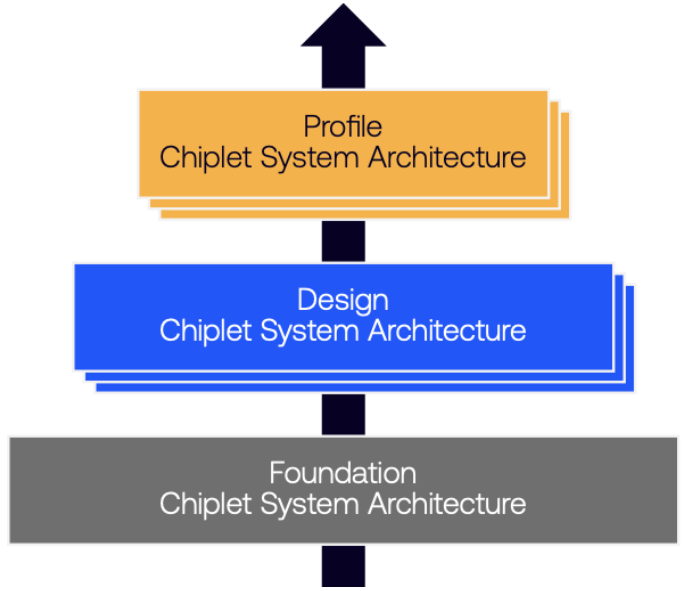
Most errors are fixed in-line by FEC; replay is the fall-back, so the latency penalty is rare.

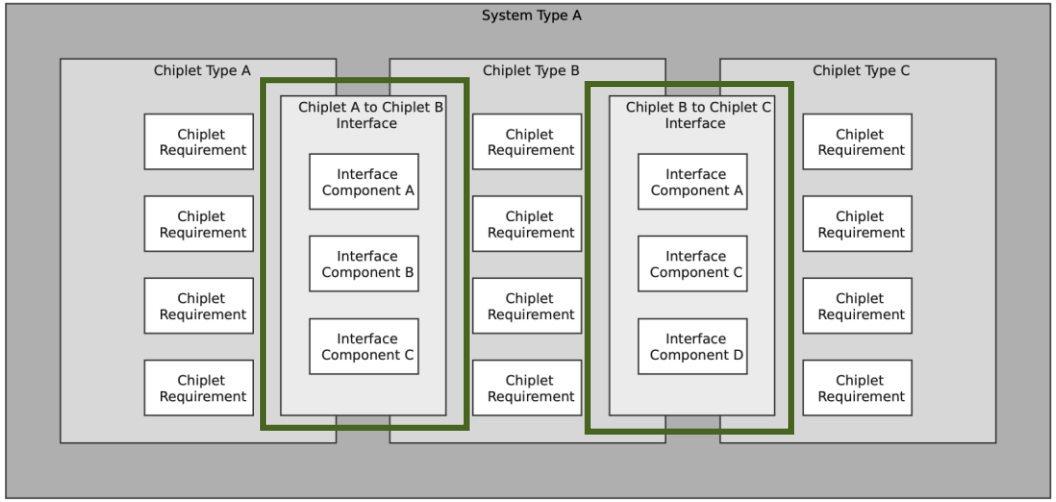
LSI Fits the ARM FCSA Quite Nicely!

- Fits layered Architecture in FCSA V1.0.0

- LSI intends to define ”Design” spec this architecture layer
  - Different products can be designed following LSI spec

- Defines the interface specification of FCSA Component Structure





Call to Action

- Join the OCE LSI Workstream
  - Participate in LSI standard enhancements through open review process
  - Critical to extend the ecosystem
- Goal to release Phase 1 spec around Nov 2026
- LSI Specification Draft
- Visit the wiki page
  - https://www.opencompute.org/w/index.php?title=Server/OpenChipletEconomy/Lightweight-Serial-Interconnect
- Mailing list: https://ocp-all.groups.io/g/ocp-lsi
- Workstream calls : Every other Tuesday 9am – 10am pacific time
  - Subscribe to calendar

Thank You!

